

High-Fidelity Multi-Physics Guidelines for Model Validation and Uncertainty Quantification



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A photograph of a nuclear reactor's internal components, showing a complex network of pipes, valves, and structural elements, illuminated with a blue light.

Reactor Dynamics and Fuel Modeling Group

Outline

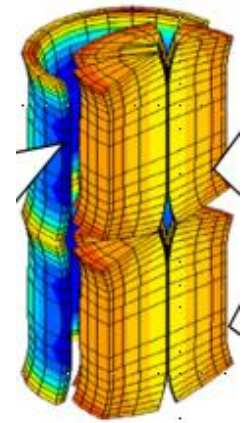
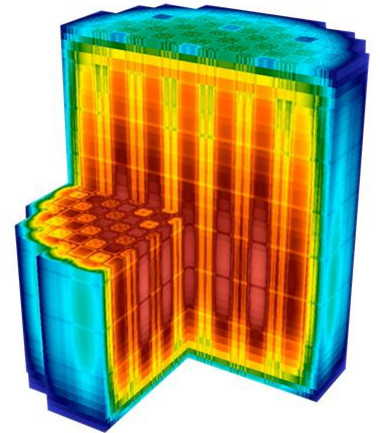
- Introduction and Background in Model Validation
- Challenges for Validation and Uncertainty Quantification
- Recommendations for Validation and Uncertainty Quantification
- Demonstration Case Study
- Conclusions

Outline

- **Introduction and Background in Model Validation**
- Challenges for Validation and Uncertainty Quantification
- Recommendations for Validation and Uncertainty Quantification
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Introduction

- Verification, Validation, and Uncertainty Quantification (VVUQ) of computational models are essential for demonstrating predictive credibility.
- The role of high-fidelity, high-resolution, modeling and simulation (M&S) keeps expanding supporting design studies for new reactors, safety analysis, and operation optimization among many other applications.
- VVUQ for high-fidelity and high-resolution models is particularly challenging due to the complexity of the models and the lack of pin-resolved experimental data.
- Recent progress has been made in the field of verification of high-fidelity multi-physics with improvements in: (1) coupling techniques, (2) convergence acceleration, (3) efficient multi-scale approaches, (4) data transfer methods, and (5) software engineering practices.
- A similar progress is needed in the field of model validation and uncertainty quantification.

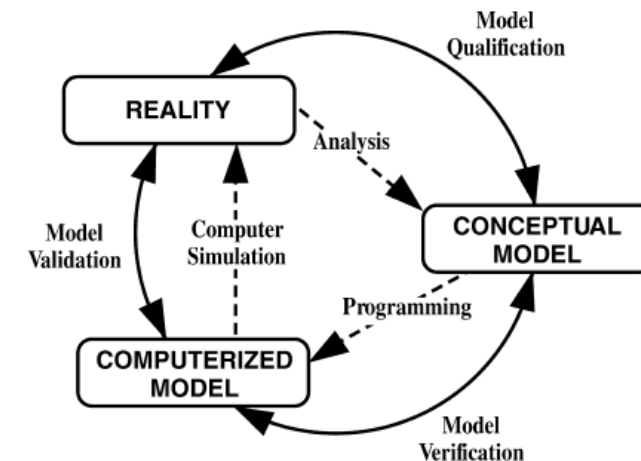


History of Model Validation – Early Era (Before 1970)

- The history of VVUQ spans many decades and engineering disciplines. It can be split in three main eras: early era (before 1970), awareness era (up to 1990), modern era (up to today).
- During the early era, some initial attempts at V&V emerged in the scientific community, but the lack of consistent terminology and definitions hindered systematic progress.
- In nuclear engineering, the design of nuclear systems relied predominantly on experimental data, from which simplified models were developed with large safety margins and strong conservatism.
- The lack of advanced M&S capabilities, as well as the limited computational resources, restricted the application of these models to support studies and analyses. Therefore, rigorous V&V was not of primary importance.
- A key contribution from this area was the recognition of the necessity of V&V for scientific models.
- Although specific methods or standards had not yet been established, important concepts were introduced, such as the usage of experimental data for model validation.

History of Model Validation – Awareness Era (Up to 1990)

- Significant rise in interest for V&V across various engineering communities.
- The construction of new nuclear reactors of increasing size led to the development of safety acceptance criteria and the first M&S for licensing purposes.
- The first systematic definitions were established for model V&V by the Society for Computer Simulations in 1979.
- Verification was defined as “**substantiation** that a computerized model represents a conceptual model within specified limits of **accuracy**”.
- Validation was defined as “**substantiation** that a computerized model, within its **domain of applicability**, possesses a satisfactory **range of accuracy** consistent with the **intended application** of the model”.
- Accuracy requirements must be defined in advance, and the model’s accuracy should be evaluated with respect to these requirements for the intended application.
- During the awareness era many approaches and techniques were proposed to answer the questions “Who should decide if the model is valid?” and “How should the validity of the model be assessed?”.



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History of Model Validation – Modern Era (Up to Today)

- Interest in V&V remains strong with solid foundations, practices, and standards being established.
- V&V is being adopted by most engineering fields with many organizations, including the U.S. Nuclear Regulatory Commission (NRC), making it a requirement for computational models.
- The V&V definitions were refined to eliminate some of the ambiguities, converging into the definitions used today by most engineering disciplines.
- Validation definition was updated to “the **process** of determining the degree to which a model is an **accurate representation** of the real world from the perspective of the **intended uses** of the model”.
- One important clarification in this definition is that both validation and verification are ongoing processes, that need to be repeated whenever: (1) there is a model update, (2) a new application is envisioned, or (3) new experimental information is available.
- Experimental observations are explicitly regarded as the best available measure of real-world phenomena. It is important to note that this does not treat the experimental data as more accurate than the model predictions.
- Uncertainty quantification is at the core of the validation process.

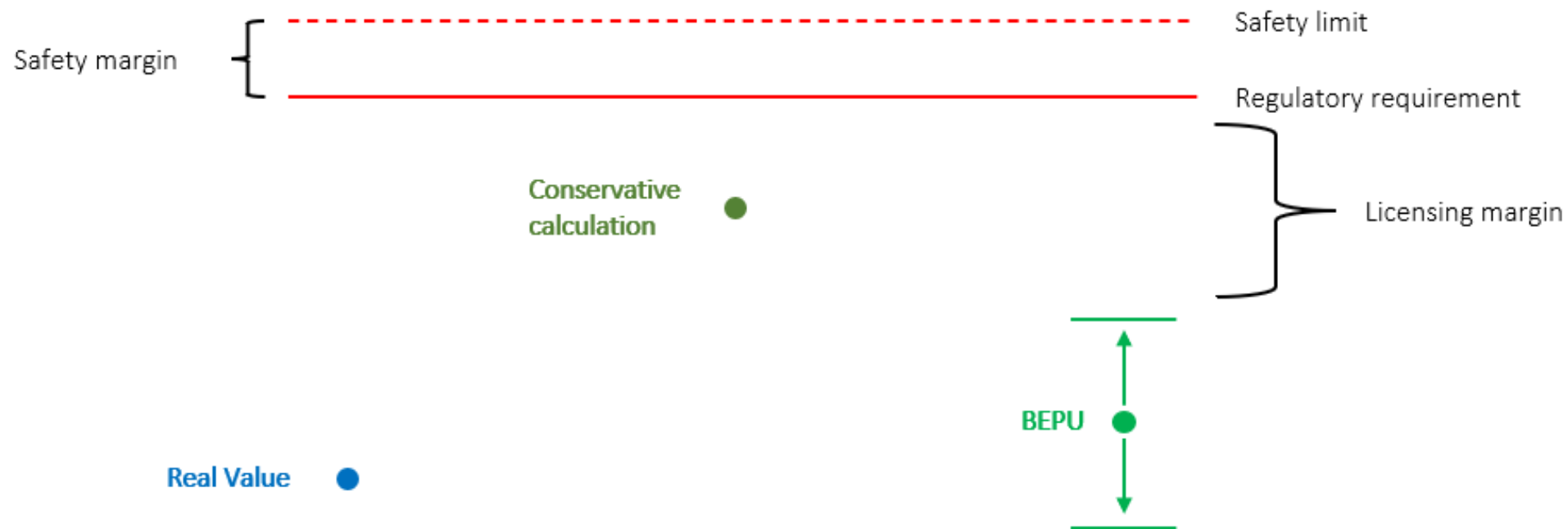
$$y_{real} + \epsilon_{exp} = y_{pred} + \epsilon_{model} + \epsilon_{num} + \epsilon_{uq}$$

Model Validation and Uncertainty Quantification

- The modeler is responsible for selecting the best-suited and feasible representation of each source of uncertainty.
- Once all these sources of uncertainty are quantified, appropriate validation metrics $V(y_{pred}, y_{exp})$ are used for measuring the model accuracy.
- To support the model validation process, a preliminary task often used is the Phenomena Identification and Ranking Table (PIRT). PIRT results in systematic identification and ranking of the importance of different physical phenomena with regards to a specific quantity of interest.
- A typical validation framework includes:
 1. **Model accuracy assessment.** Comparison of model predictions against experimental data using appropriate validation metrics.
 2. **Model predictive capability.** Model prediction on the intended application conditions. This can be an interpolation within the validation domain or an extrapolation outside of this domain. The interpolation or extrapolation does not only involve the model prediction itself, but also the associated model uncertainty. The outcome of this step is the total uncertainty of the model and the corresponding accuracy at the application conditions.
 3. **Model adequacy assessment.** The accuracy is compared against the accuracy requirements for the intended application to decide if the model is adequate.

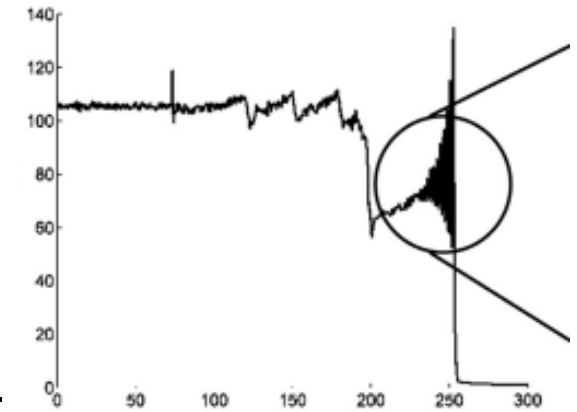
Multi-Physics Model Validation Activities in Nuclear Engineering 1/3

- Historically, model validation in nuclear engineering has been a core part of safety analyses.
- Initially, conservative approaches were employed to assess safety margins.
- As computational resources increased and M&S fidelity improved, Best Estimate Plus Uncertainty (BEPU) approaches were introduced, with VVUQ one of its key elements.
- Examples of such approaches are CSAU and EMDAP by U.S. NRC.
- The Wilks' method is often employed to estimate the 95% tolerance limit with 95% confidence.



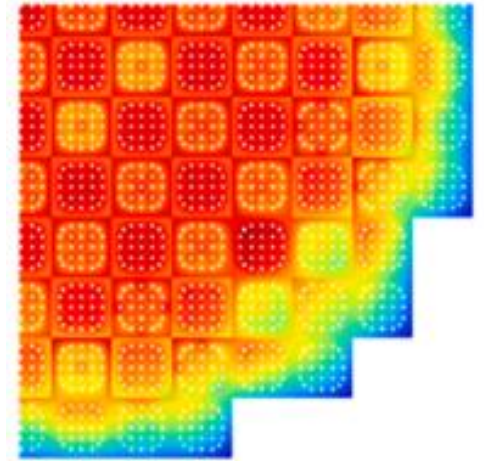
Multi-Physics Model Validation Activities in Nuclear Engineering 2/3

- Many multi-physics validation benchmark activities have been conducted by the nuclear M&S community in the past three decades.
- Initially, the benchmarks focused on the validation of traditional multi-physics tools emphasizing code-to-code comparisons and integral experimental data
- Examples of benchmarks developed for these tools include the OECD/NRC Pressurized Water Reactor Main Steam Line Break (PWR-MSLB), the OECD/NRC Boiling Water Reactor Turbine Trip (BWR-TT), and the OECD/DOE/CEA VVER-1000 Coolant Transient (V1000T).
- A multi-level approach is applied through different benchmark Phases that define exercises of progressively increasing modeling complexity.
- The next generation of benchmarks had higher quality experimental data up to the assembly level. Examples include the OECD/NEA Kalinin-3 VVER-100 and OECD/NRC Oskarshamn-2 BWR stability benchmark.
- More recently, benchmarks targeting also novel high-fidelity, high-resolution multi-physics tools have been developed. Uncertainty analysis and high to low fidelity methods are also assessed.
- Examples include the OECD/NEA LWR-UAM, MPCMIV, and TVA Watts Bar Unit 1.



Multi-Physics Model Validation Activities in Nuclear Engineering 3/3

- The third category of validation activities is related to the validation of high-fidelity, high-resolution multi-physics tools.
- One of the most prominent examples is the VERA code suite developed under the Consortium for the Advanced Simulation of LWRs (CASL).
- VERA has a pin-by-pin resolution using, among others, the MPACT code for core neutronics, CTF for subchannel thermal-hydraulics, and Bison for fuel performance.
- The U.S. Department of Energy (DOE) Nuclear Energy Advanced Modeling and Simulation (NEAMS) program included validation activities for advanced reactors using MOOSE
- An example is the multi-physics transient modeling of heat pipe and gas-cooled microreactors using a coupling between Griffin and Bison.
- Recently, the Cardinal wrapper allow the capability to couple MOOSE tools with with the Monte Carlo neutronics code OpenMC and the computational fluid dynamics (CFD) code NekRS. NekRS-Bison coupled calculations were performed and compared against experimental data.
- In Europe, the nTRACER 2D/1D planar method of characteristics (MOC) neutronics code is coupled with the CTF subchannel and validated using the OECD/NEA TVA-WB1 benchmark.
- Validation activities using a coupling of Serpent, SUBCHANFLOW, and TRANSURANUS have been performed for PWR cycle depletion (McSAFE).



Outline

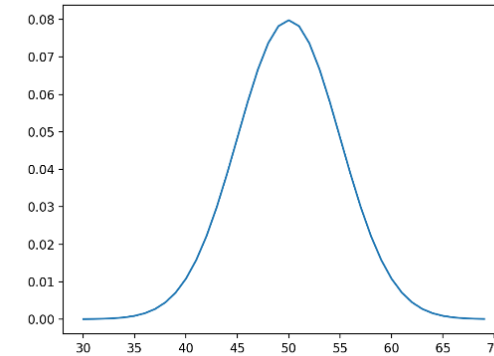
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Input and Output Dimensionality

- High-fidelity multi-physics models involve a vast number of input parameters, ranging from nuclear data to thermal-hydraulics closure models and fuel behavior models.
- Identifying all the relevant inputs and characterizing their uncertainties is challenging, particularly since some of these inputs might exhibit strong correlations.
- The large input dimensionality also typically increases the number of code evaluations for uncertainty quantification and sensitivity analysis, leading to a stronger reliance on surrogate models. The development of accurate surrogate models (e.g., training, validation, testing) becomes more complicated and time-consuming in very large input spaces.
- Most of the multi-physics models' predictions are multivariate outputs in space and time, leading to strong spatiotemporal correlations. There could also be strong correlations between different outputs. For example, between the linear power and fuel temperature.
- If there are available experimental data for these multivariate outputs, then the typical univariate validation metrics will not be appropriate. Validation metrics dedicated to multivariate output should be used instead.
- Multivariate outputs also pose challenges in the development of surrogate models for UQ and sensitivity analysis (SA). Various techniques can be used to reduce the output space by finding a latent space of lower dimensionality.

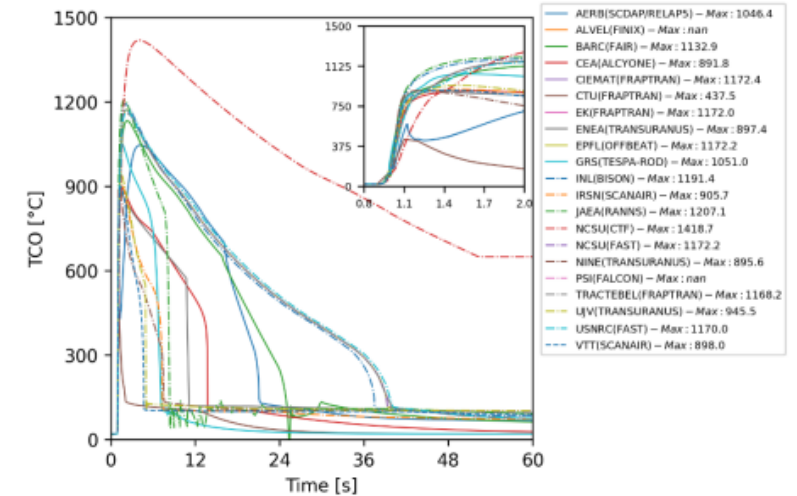
Input Uncertainty Quantification

- There are two general types of uncertainties:
 1. Aleatoric: inherent variability (irreducible)
 2. Epistemic: due to lack of knowledge (reducible)
- Separating epistemic from aleatory uncertainty requires a comprehensive approach that is seldom done in practice for nuclear engineering applications, due to the lack of detailed uncertainty information and to the increased computational cost for propagating these uncertainties.
- Inputs related to inlet conditions (e.g., mass flow rate) and manufacturing (e.g., fuel dimensions) are usually characterized as aleatory with a normal or uniform distributions.
- The uncertainty of empirical or semi-empirical modules is obtained by comparison against selected experimental measurements.
- In some cases, inverse uncertainty quantification and calibration techniques are used to obtain the modules input parameters uncertainty quantification.
- Care should be taken not to include data from model calibration into model validation.



Model Complexity

- High-fidelity multi-physics models usually include:
 - non-linearities,
 - discontinuities,
 - strong interactions between the uncertain inputs.
- Such complex behaviors practically necessitate the usage of sample-based stochastic approaches for uncertainty propagation that result in multiple code evaluations.
- Due to the high associated computational cost, a limited number of samples can be generated, which restricts the applicability of various UQ and SA methods.
- The coupling convergence can become challenging.
- It is important to have a parsimonious model, meaning that it is as simple as possible to model the important phenomena of the intended application and available experimental measurements.



Consistent Input Sampling

- In stochastic approaches for uncertainty propagation, random samples are generated for the input parameters.

- For some inputs, consistency rules need to be applied at each sample.

$$\sigma_t = \sigma_f + \sigma_s + \sigma_\gamma + \dots$$

$$r_{co} = r_f + t_{gap} + t_{clad}$$

- Sometimes the consistency requirement can be more subjective. For example, the perturbation of the fuel rod radius in a reactor core.
- One approach is to consider the same perturbation for all fuel rods. This assumes that all the fuel rod dimensions are fully correlated leading to essentially only one random input.
- Another approach is to consider that all perturbations are independent for each fuel rod or fuel assembly. This results in a significantly larger number of uncertain inputs.
- Both methods are valid, however, in practice, the former approach is used more often due to its simplicity in applying the input perturbations. The latter approach is more accurate when the underlying uncertainties represent random effects, but its implementation might not be feasible if the individual fuel rods are not explicitly modeled.

Consistent and Comprehensive Uncertainty Propagation

- Double-counting uncertainties between the coupled physics codes must be avoided.
- In the literature, UQ frameworks for addressing this aspect were proposed, where the main idea is to apply the same samples to all common sources of uncertainties within each code.
- A more complicated situation arises when model parameters depend on fundamental sources of uncertainties implicitly.
- The treatment of appropriate reactor operational procedures for each input sample is also important. One approach is to fix the operational procedures (e.g., control rod positions, total power) and compute the associated output's uncertainty.
- A different approach would be for each sample to adapt the operation procedures to maintain criticality or any other operational target. Both approaches are valid, but it is important to acknowledge the assumptions involved with the selected approach and draw the appropriate conclusions from the UQ studies.
- In principle, the epistemic, aleatory, and mixed uncertainties should be treated separately. All possible sources of uncertainty, including numerical uncertainty and stochastic sampling uncertainty should be considered. This is practically not possible in most cases.

Quantitative Sensitivity Analysis

- The goal is to determine how the inputs influence the outputs for screening/quantitative (e.g. small/medium/large) or qualitative (e.g. 20%) purposes:
 - Understand the behavior of the model.
 - Reduce input dimensions to simplify the model's complexity.
 - Identify the most important inputs to focus on reducing their uncertainties.
 - Measure the impact of statistical assumptions (e.g. independence, sample size, pdf).
- We can identify 4 types of input contributions to output's uncertainty:
 1. Direct effect through the model: linear ($\partial Y/\partial X$) or non-linear.
 2. Indirect effect through interactions with other inputs.
 3. Variance of the input.
 4. Correlation of the input with other inputs.
- There are various approaches that can quantify some of these contributions under various approximations.
- The most detailed methods are the ANOVA- Analysis of Variance (e.g. Sobol) methods that decompose the output's variance to input contributions.

Scarce Experimental Data

- There are few historical data that can be used for high-fidelity multi-physics model validation, combined with some relatively new validation experiments.
- This issue is even more pronounced for high-resolution data and for advanced reactors.
- The uncertainty information is limited.
- The experiments are usually not repeated to estimate the random error. An estimate of the measurement uncertainty in the form of standard deviation is sometimes provided.
- A common choice is to consider the uncertainty as aleatory and use a normal distribution with the provided standard deviation centered on the measured value.

Assessment of Model Predictive Capability

- Assessment of model predictive capability can be very complicated and poses many challenges. The application domain can be in an interpolation or an extrapolation region of the validation domain covered by the validation experiments.
- The transferring or scaling of the estimated accuracy from the validation domain to the application domain is not straightforward and depends on the underlying physical phenomena.
- In nuclear reactor M&S, it is anticipated that most of the application domains will lie in the extrapolation regions due to the difficulty in performing large-scale transient experiments.
- Thermal-hydraulics is the physics domain that has historically focused more on the scaling distortions between the large-scale nuclear reactor application and the scaled-down selected experiments that aim to represent most of the important physical phenomena.
- Non-dimensional parameters are typically used, and various scaling methods have been proposed from simple linear scaling to complex hierarchical procedures.
- Similarity indices can also be used to measure the degree of similarity between the various experiments and the intended application.
- More research is needed for investigating the scaling of high-fidelity multi-physics model predictions.

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Input and Output Dimensionality

- When there are strong correlations, linear and non-linear dimensionality reduction techniques can be very powerful. These methods transform the original high-dimensional space into a lower-dimensional latent space.
- Common linear approaches that have been used in nuclear engineering are PCA or proper orthogonal decomposition (POD), and dynamic mode decomposition (DMD).
- For non-linear complex behaviors, neural network-based methods such as autoencoders can also be used.
- A different approach to reduce the input space dimensions involves identifying and discarding a subset of non-influential inputs for a specific output of interest. This is called screening.

	Dimensionality Reduction	Screening
Examples	PCA/POD, DMD, DEIM, Autoencoders	OAT, Morris, HSIC
Applicability	Transforms high-dimensional space to low-dimensional latent space leveraging correlations and patterns.	Identifies and discards a subset of non-influential inputs.
Pros	+ unsupervised + significant dimensionality reduction when strong correlations are present + applies to both input and output spaces	+ measures impact of inputs on output accounting for complex behaviors + does not depend on the input correlations
Cons	- not easily interpretable latent space - loses efficiency if there are no strong correlations - does not account for the impact of the input to the output	- applied for a specific output of interest - computationally expensive for large input dimensions

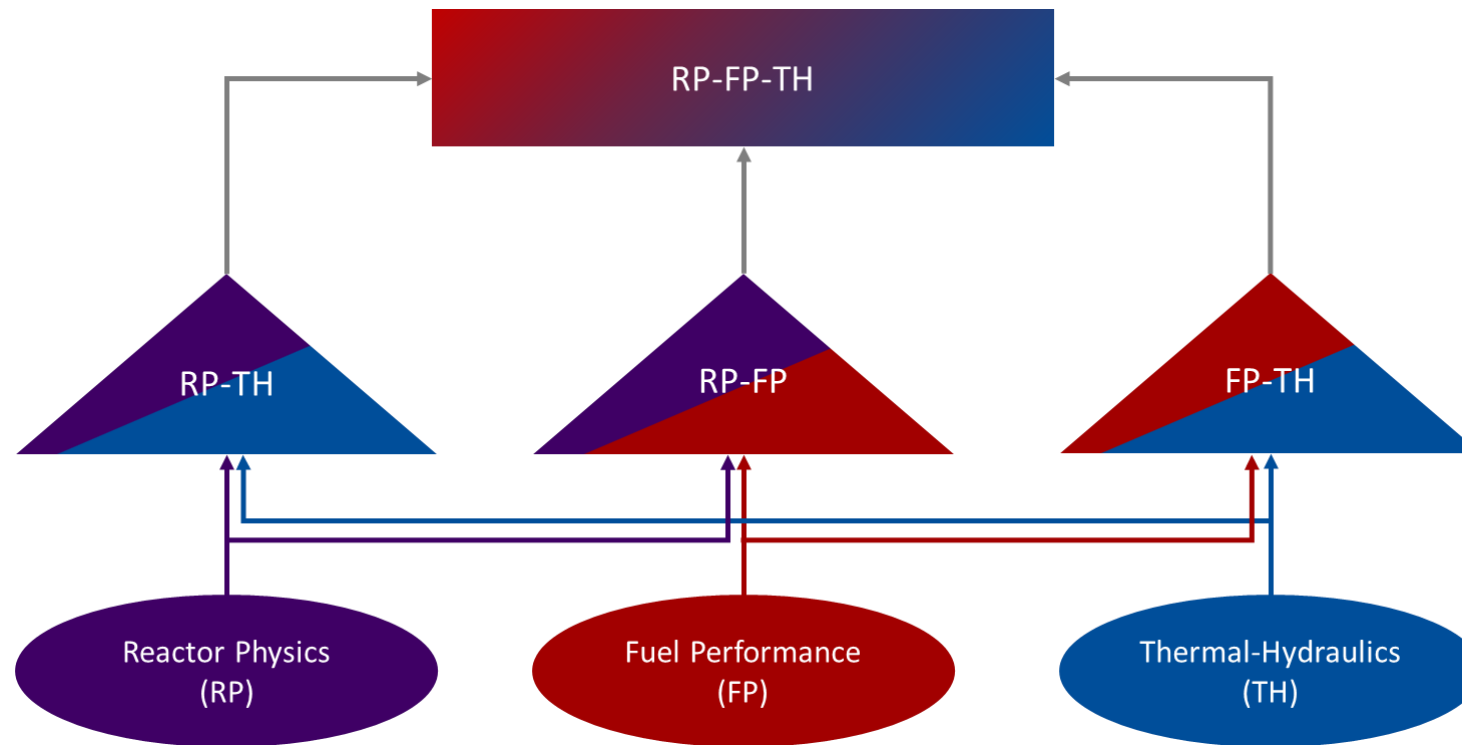
Input Uncertainty Quantification

- Many approaches from reported manufactured uncertainty and up to posterior distributions obtained via Bayesian calibration on some selected experimental data.
- Comprehensive treatment of epistemic and aleatory uncertainties is computationally expensive, since it involves a nested sampling loop.
- For this reason, most multi-physics UQ studies treat all inputs as aleatory.
- However, it is suggested to provide a general characterization, as performed in the Table, acknowledging their different nature, and then proceed by stating the assumption of not separating them for practical purposes.

Input Category	Example	Suggested Uncertainty Type
Initial/Boundary conditions	inlet mass flow rate	aleatory
Manufacturing parameters	fuel diameter	aleatory
Microscopic cross-sections	U-235 fission	aleatory
Material properties	fuel thermal conductivity	aleatory
Model parameters	cladding creep	epistemic

Model Complexity 1/3

- A multi-level hierarchical approach is of strong interest to progressively increase the level of coupling and the associated model validation.
- A satisfactory coupling level can be selected for performing the uncertainty quantification that is cost-effective.



Model Complexity 2/3

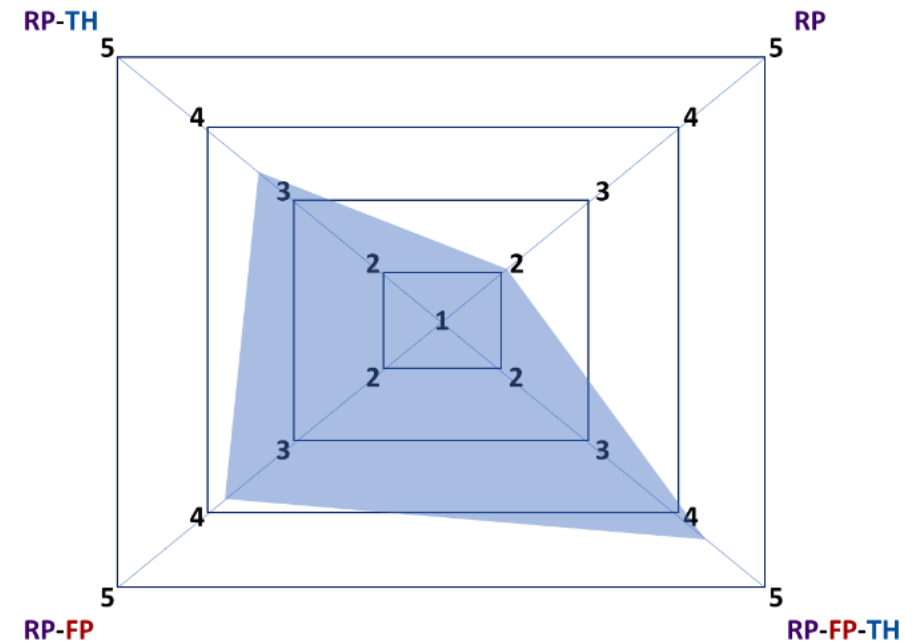
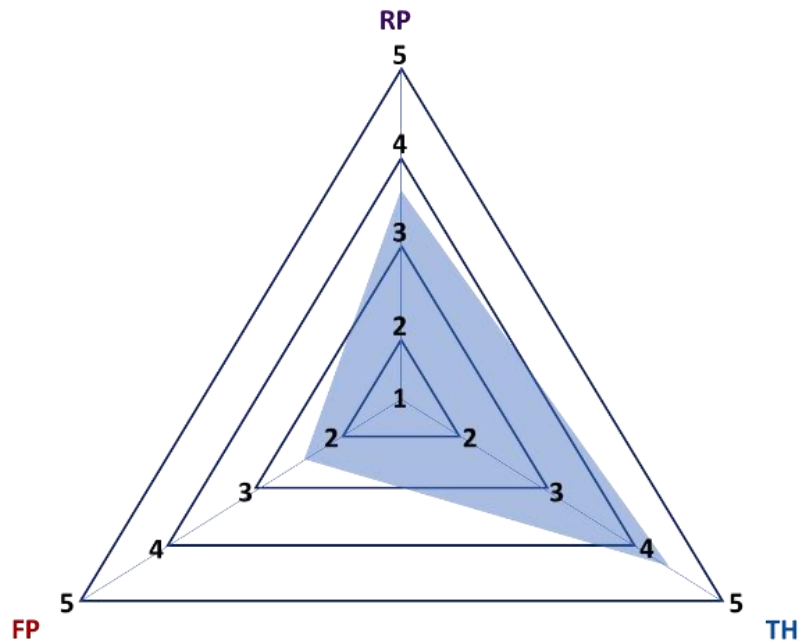
- A model scoring approach is suggested to easily compare the performance of each coupled model against the available experimental data.

$$s_i = \frac{N}{\sum_{q=1}^N \frac{1}{s_i^q}}, \quad s_2^1 \begin{cases} 1: \delta_2^1 > 500 \text{ pcm} \\ 2: \delta_2^1 = 350 \text{ pcm} \\ 3: \delta_2^1 = 200 \text{ pcm} \\ 4: \delta_2^1 = 100 \text{ pcm} \\ 5: \delta_2^1 < 50 \text{ pcm} \end{cases}$$

- These scores are computed for each measured quantity of each physical domain.
- It is essential to acknowledge the inherent subjectivity of these scores, which arises from the chosen accuracy requirements.
- Whenever possible, regulatory criteria should be prioritized, followed by established engineering standards. In the absence of such information, expert judgement may be applied.
- In any case, a justification should be provided for the selected accuracy requirements with level 1 indicating an excellent performance, while level 5 indicating an unacceptable performance. Levels 2-5 should provide sufficient granularity to distinguish between different models.

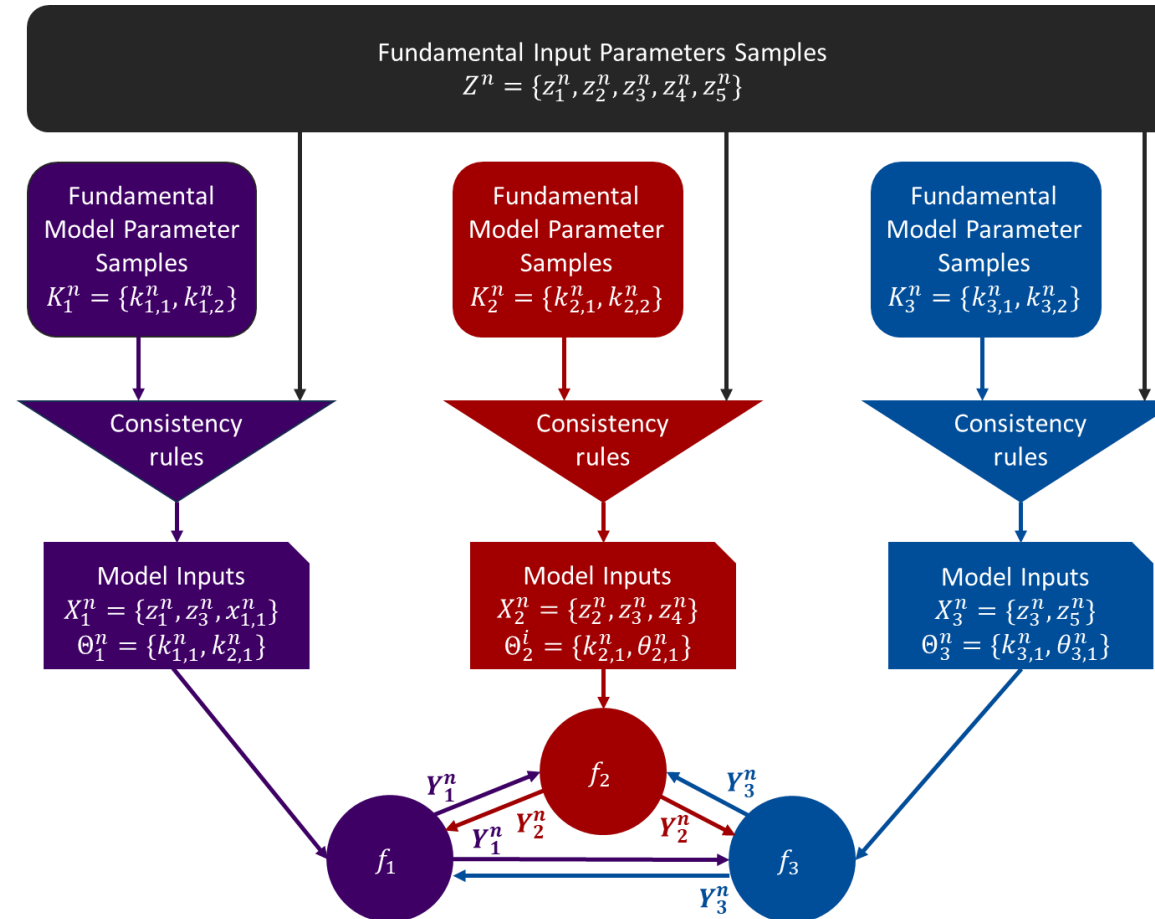
Model Complexity 3/3

- The scores can be used to create two illustrative scoring diagrams.
- In the first type of diagram, the scores of each physics domain are drawn for a coupling between RP, FP, and TH.
- In the second type of diagram, the scores for one quantity of interest are drawn for different coupling levels.



Consistent Sample-Based Uncertainty Propagation 1/2

- We propose the usage of sample-processing diagrams for uncertainty propagation to identify and track all the interactions between the fundamental samples and record the required consistency rules.



Consistent Sample-Based Uncertainty Propagation 2/2

- The complicated nature of many of the inputs (e.g., cross-sections) and their interactions through all the coupled codes might make it challenging to perform on-the-fly sampling.
- An offline sampling approach is recommended in this case, where many input samples are generated in advance and stored.
- This is used by the SCALE code suite, where a set of 1000 pre-computed microscopic cross-section samples are provided with the code suite. Several benefits stem from such approaches:
 - **Modular workload.** One team can focus on the stochastic sampling generation and make sure that all consistency rules are met, while another team can work on the coupled models.
 - **Repeatability.** The calculations can be repeated in case of a code crashed, or if calculation was stopped.
 - **Model comparisons.** The performance of two different models can be compared on a sample-by-sample basis to better understand the discrepancies.
 - **Similarity studies.** If the same input samples are applied to different applications, it can allow the estimation of similarity indices between these applications.
 - **Sensitivity analysis.** Trends and behaviors between the inputs and outputs can be used to infer some simple sensitivity analysis conclusions.

Sensitivity Analysis

- For most UQ and SA multi-physics studies, a realistic number of code evaluations is in the range of 100-1000. Based on this constraint, two different categories of sensitivity analysis approaches are suggested.
- The first category consists of methods that rely only on code evaluations and thus are restricted to a limited number of samples.
 - Pearson correlation that measures the linear relationship between input and output.
 - Spearman correlation that allows for a monotonic relationship between input and output.
 - Dependence measures that allow non-linear relationships between inputs and outputs.
 - Johnson indices have recently been studied with promising results as an efficient variance decomposition method.
- The second category of methods provide a more detailed and quantitative breakdown of the importance of each input, but that consequently require significantly more code evaluations, something not feasible using only the multi-physics model (e.g. Sobol, Shapley indices).
- For such sensitivity analysis studies, the construction of a surrogate model is necessary. The surrogate model can have any architecture from polynomial functions to gaussian processes, and up to neural networks and random forests.

Multivariate Outputs

- Consistent multivariate validation metrics based on U-pooling technique exist in literature but are significantly more difficult to implement and interpret. More research is needed.
- Simple not necessarily consistent metrics can be used that target local, regional, and global aspects. Also, feature based metrics could be considered. If we use a temporal output as an example:
 - Local metrics will measure the discrepancies at each point in time.
 - Regional metrics will measure the discrepancies for identified regions of interest.
 - Global metrics will aggregate the discrepancies across all time steps into one representative value.
 - Feature metrics target discrepancies in features such as the pulse width in a rod ejection accident.
- Extension to any phase-space is straightforward using weights that depend on the nature of the predicted quantity (e.g. point estimate, volumetric average).

$$\delta = \sum_{i=1}^N |y_i - e_i| \frac{w_i}{w}, \quad w = \sum_i^N w_i$$

Assessment of Model Predictive Capability

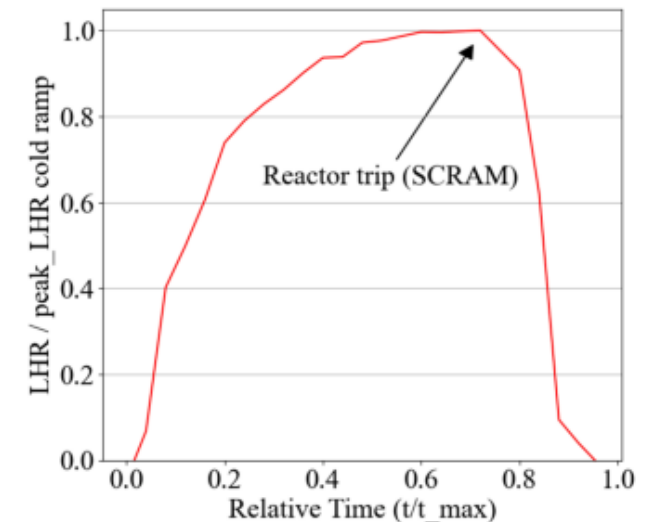
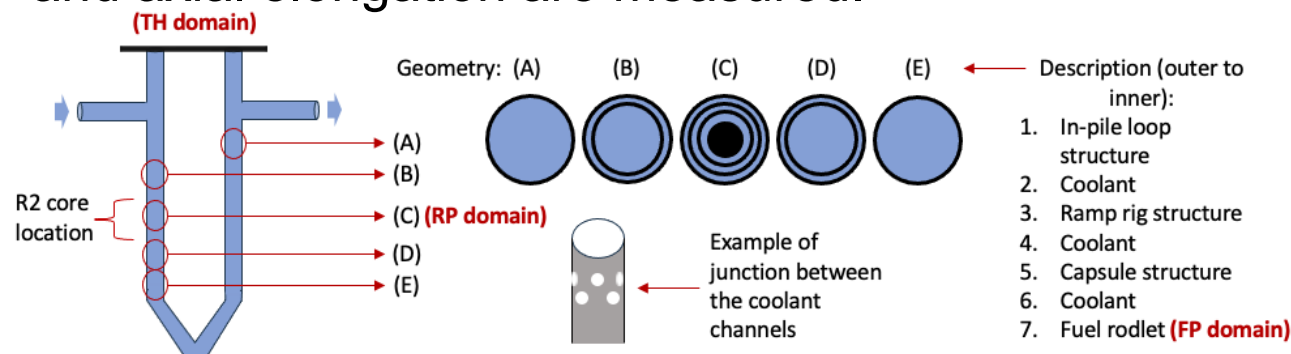
- Computational cost and model complexity limit the applicability of various existing methods.
- A first category of approaches is the interpolation/extrapolation of model accuracy based on some identified features following a PIRT. T
- The features determine the dimensionality of the validation and application domain, and the interpolation/extrapolation can be performed with different techniques (e.g., linear).
- A second category of approaches can leverage the estimation of stochastic similarity indices between validation experiments and the application of interest.
- An interesting approach, which is related to similarity indices, is the Physics-guided Covered Mapping (PCM).
- The main required functionality is the ability to apply the same samples across all models for the validation experiments and the application. Additionally, the results obtained on the validation experiments can be reused for multiple applications of interest without additional code evaluations.

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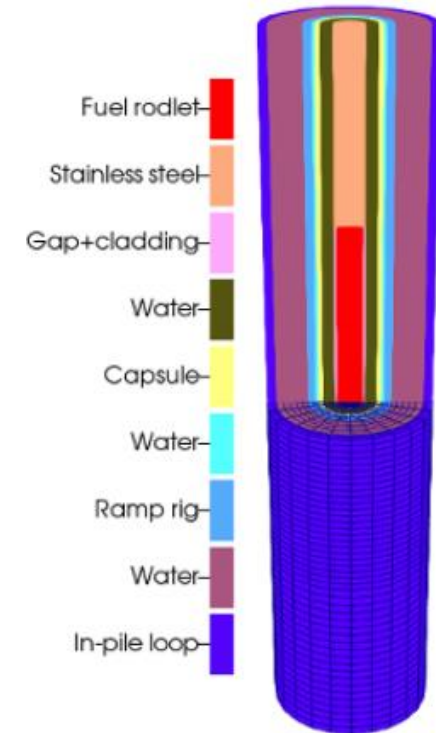
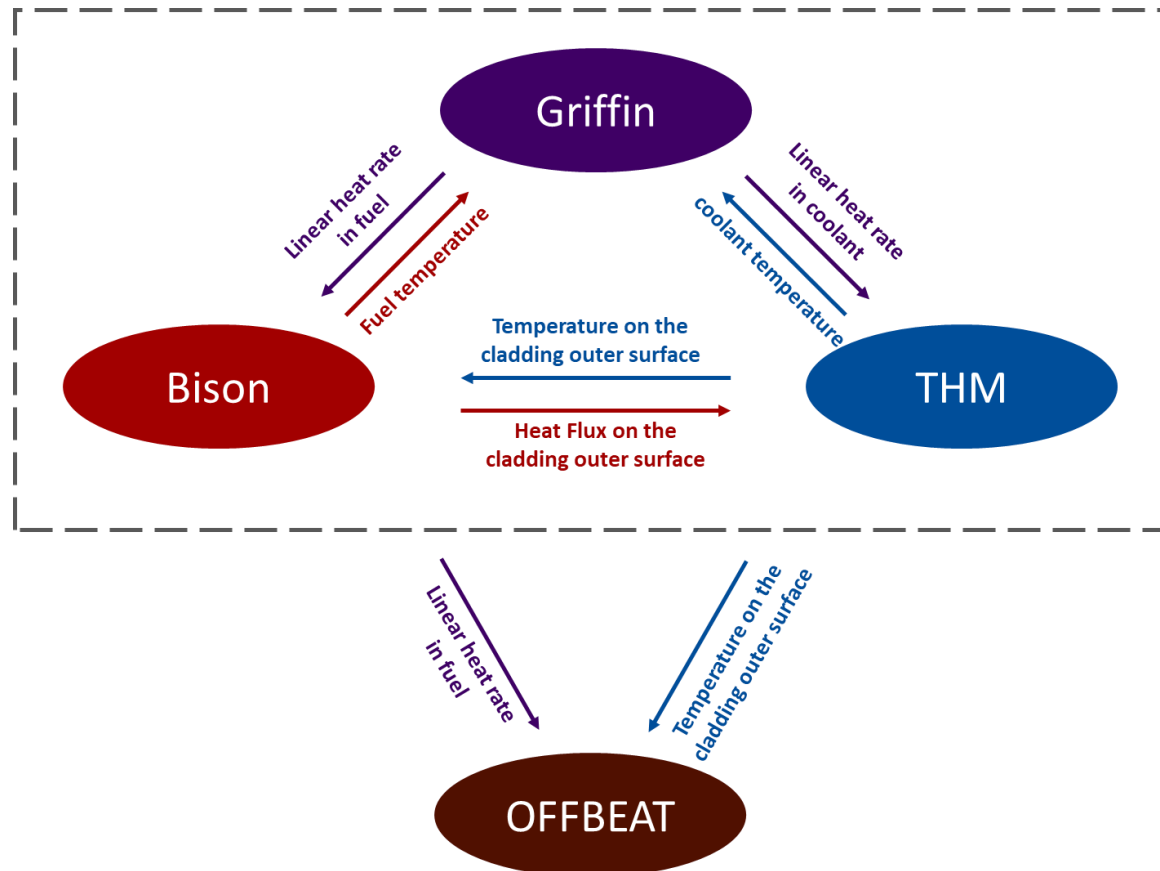
MPCMIV Benchmark

- Some of the presented guidelines are demonstrated using the MPCMIV benchmark.
- The main goal of the MPCMIV benchmark is to provide validation data for transient multi-physics M&S that apply to both traditional and novel multi-physics tools.
- Experimental data from a cold ramp test conducted in the Studsvik R2 reactor. The test is carried out by inserting the fuel rodlet from the top into the in-pile loop irradiation position. The fuel rodlet is irradiated at a constant power for less than one minute before the reactor is shutdown.
- The fuel rodlet was refabricated from a father rod previously irradiated in a Boiling Water Reactor (BWR) for three cycles.
- For RP, the power is estimated based on a calorimetric technique. For TH, the rod inlet/outlet temperature and mass flow rates are measured. For FP, the cladding outer diameter and axial elongation are measured.



Multi-Physics Model

- Griffin SN transport solver is coupled to Bison fuel performance code and THM simplified 1D thermal-hydraulic code.
- Due to convergence issues in Bison contact model, a one-way coupling with OFFBEAT fuel performance code is used.
- Only the fuel rodlet domain is modelled for both base irradiation and the cold ramp test.



Multi-Level Validation and Scoring Approach

- From single-physics, to two-physics coupling, and up to full coupling is studied.
- The following measured quantities are used:
 - Cladding axial elongation (ϵ_{clad}).
 - Inlet temperature of the ramp rig (T_{rin}).
 - Outlet temperature of the ramp rig (T_{rout}).
 - Outlet temperature of the in-pile loop (T_{lout}).
- The local metric is simply the absolute error at the instance of the measured peak value.
- Two regional metrics are computed: region 1 for the first 2/3 of the cold ramp, capturing the rapid transient effects, and region 2 for the last 1/3 of the transient, capturing the slower transient effects as the steady state is approached. For these regional metrics, the temporal mean and standard deviation of the absolute error are provided.
- For the global metric, like the regional metrics, the mean and standard deviation of the absolute error are computed over the entire transient.

$$s_1^q \begin{cases} 1: & \delta_1^q > 45 \% \\ 2: & \delta_1^q = 35 \% \\ 3: & \delta_1^q = 25 \% \\ 4: & \delta_1^q = 15 \% \\ 5: & \delta_1^q < 5 \% \end{cases}$$

$$s_2^q \begin{cases} 1: & \delta_2^q > 1.25 K \\ 2: & \delta_2^q = 1.00 K \\ 3: & \delta_2^q = 0.75 K \\ 4: & \delta_2^q = 0.50 K \\ 5: & \delta_2^q < 0.25 K \end{cases}$$

Results – FP Single-Physics

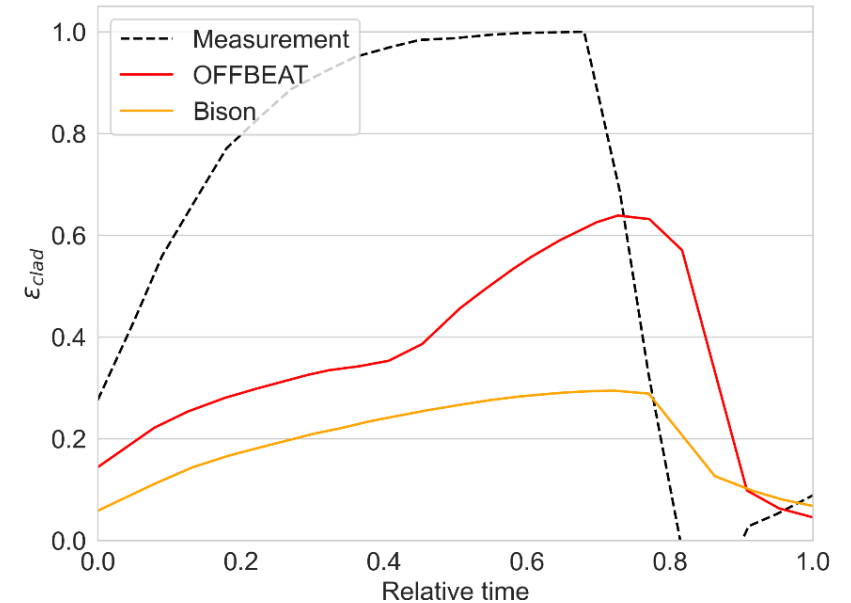
- Both OFFBEAT and Bison show poor agreement with validation score 1.
- The OFFBEAT results are slightly better resulting in improved validation metrics.

Bison

ϵ_{clad}	Validation Metric	Validation Score
Local maximum	0.71	FP: 1.0
Region 1	(0.58,0.18)	FP: 1.0
Region 2	(0.72, 0.006)	FP: 1.0
Global	(0.62,0.16)	FP: 1.0

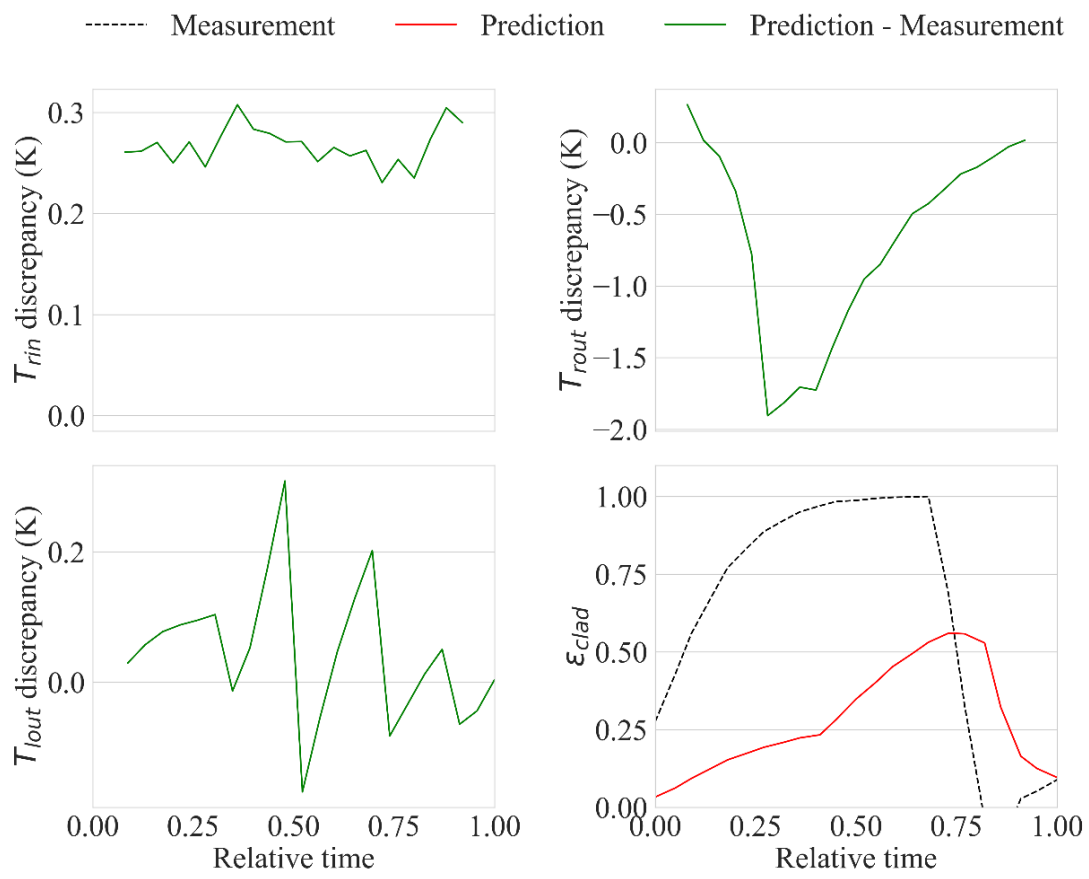
OFFBEAT

ϵ_{clad}	Validation Metric	Validation Score
Local maximum	0.38	FP: 1.7
Region 1	(0.47,0.17)	FP: 1.0
Region 2	(0.46, 0.06)	FP: 1.0
Global	(0.47,0.14)	FP: 1.0



Results – FP-TH Coupling

- Small discrepancies in coolant temperatures but still large discrepancies in fuel cladding elongation.

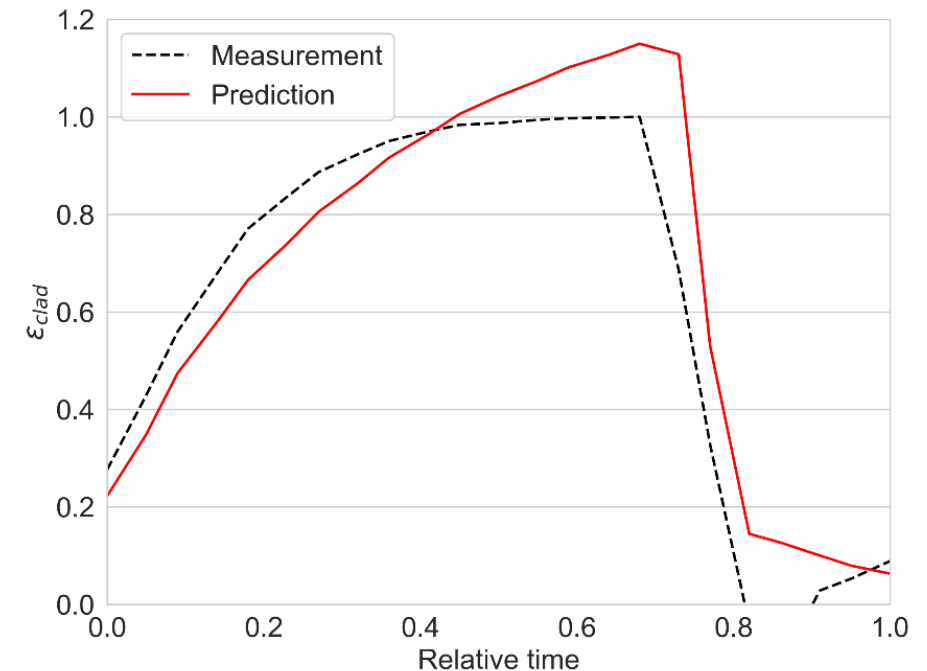


ϵ_{clad} T_{rin} T_{rout} T_{lout}	Validation Metric	Validation Score
Local maximum	0.47 0.29 0.02 0.00	FP: 1.00 TH: 4.95
Region 1	(0.59,0.16) (0.29,0.02) (1.05,0.55) (0.12,0.09)	FP: 1.00 TH: 3.11
Region 2	(0.55,0.07) (0.27,0.03) (0.11,0.09) (0.04,0.03)	FP: 1.00 TH: 4.97
Global	(0.58,0.14) (0.27,0.02) (0.76,0.64) (0.09,0.08)	FP: 1.00 TH: 4.05

Results – RP-FP Coupling

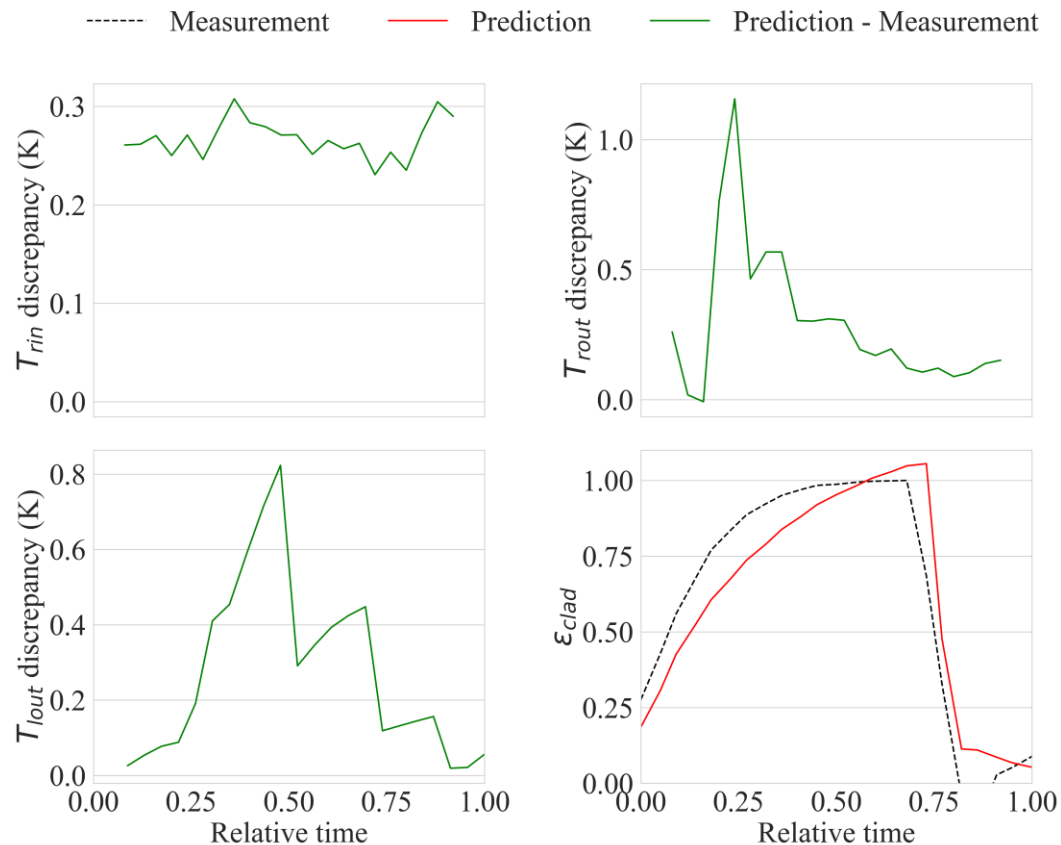
- A notable improvement is obtained for this multi-physics coupling, with the validation score ranging between 4.0 and 4.8.
- This improvement is attributed to the prediction of the LHR by Griffin using the benchmark data for the surrounding reactor environment.
- The LHR measured using the calorimetric technique is based on the coolant temperature difference between the inlet and the outlet, which includes a heating delay.
- The slower increase of the measured LHR leads to delayed contact between the fuel and the cladding and consequently in an underprediction of the cladding axial elongation.

ϵ_{clad}	Validation Metric	Validation Score
Local maximum	0.15	FP: 4.00
Region 1	(0.07,0.03)	FP: 4.80
Region 2	(0.10, 0.04)	FP: 4.50
Global	(0.08,0.04)	FP: 4.70



Results – RP-FP-TH Coupling

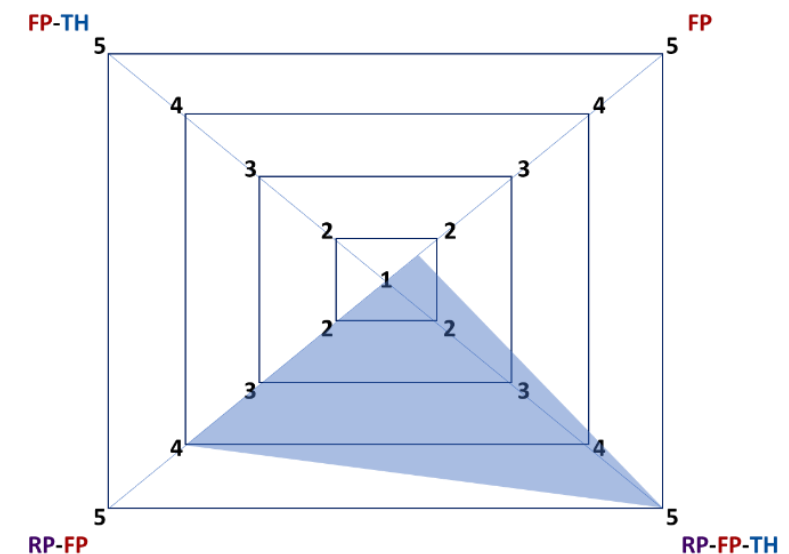
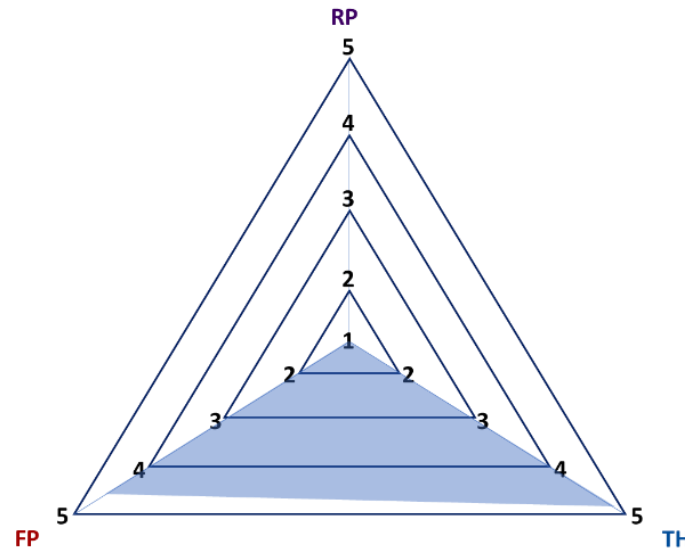
- A very good performance is obtained for both the cladding axial elongation and the coolant temperatures, with validation scores ranging between 4.2 and 5.0.



ϵ_{clad} T_{rin} T_{rout} T_{lout}	Validation Metric	Validation Score
Local maximum	0.05 0.29 0.15 0.06	FP: 5.00 TH: 4.95
Region 1	(0.13,0.03) (0.27,0.02) (0.29,0.16) (0.43,0.22)	FP: 4.20 TH: 4.66
Region 2	(0.03,0.02) (0.27,0.03) (0.12,0.03) (0.08,0.07)	FP: 5.00 TH: 4.97
Global	(0.09,0.05) (0.27,0.02) (0.23,0.16) (0.32,0.25)	FP: 4.60 TH: 4.88

Results – Multi-Level Validation

- The multi-physics model is able to predict both FP and TH physics domains quantities with excellent agreement.
- It is important to note, that in this demonstration case, FP-related phenomena dominated over TH phenomena, rendering the accurate prediction of the cladding axial elongation significantly more challenging than the coolant temperatures.
- It can be concluded that the RP-FP coupling level is accurate enough to perform the uncertainty quantification for the cladding axial elongation reducing the coupling complexity and computational cost.



Outline

- Introduction and Background in Model Validation
- Challenges for Validation and Uncertainty Quantification
- Recommendations for Validation and Uncertainty Quantification
- Demonstration Case Study
- **Conclusions**

Conclusions

- The predictive credibility of novel high-fidelity, high-resolution tools relies on appropriate VVUQ practices, which differ significantly from those used for traditional M&S tools. Many challenges exist.
- For the large number of inputs and outputs, dimensionality reduction and screening techniques are suggested.
- To address modeling complexity, a multi-level validation hierarchy is recommended that progressively increases the level of coupling among the physics domains.
- A validation scoring approach is introduced to compare results across coupling levels and identify gaps in modeling and experimental data coverage.
- Regarding the computational cost, surrogate models can be used, but they require the estimation of an additional model uncertainty.
- To ensure consistent uncertainty propagation, sample-processing diagrams are proposed to prevent sampling inconsistencies among multiple inputs.
- For the validation of multivariate outputs, local, regional, and global univariate metrics are proposed.

References

1. G. K. Delipei, Q. Faure, M. Avramova, K. Ivanov, “High-Fidelity Multi-Physics Guidelines for Model Validation and Uncertainty Quantification”, Frontiers in Nuclear Engineering, Volume 4, 2026.

Thank You



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